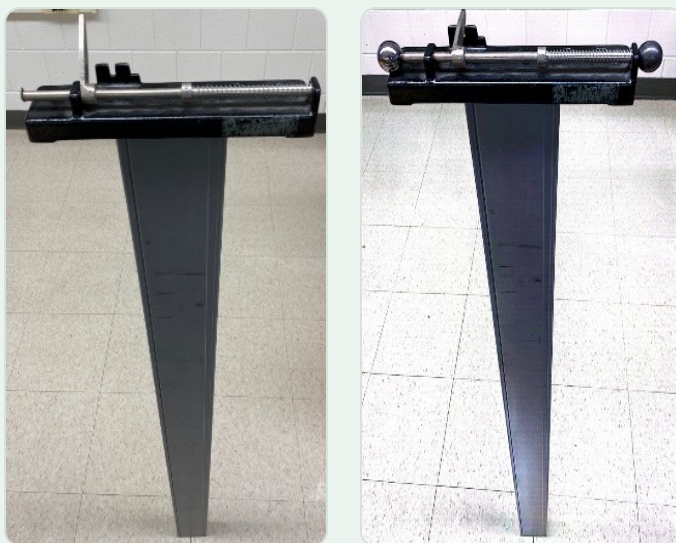


Freefall vs. Horizontal Launch

SYNOPSIS

Picture a cannonball fired from the top of a tower — the cannon aimed perfectly horizontally. At the exact instant it fires, a second cannonball is simply dropped from the same spot. The fired ball travels some horizontal distance before it lands; the dropped ball falls straight down. If you timed both balls, would one hit the ground before the other — and if so, which one — or would they land at the very same instant? This note works through a Claim–Evidence–Reasoning (CER) argument to find out, then tests it against a real demonstration.



The launcher-drop device: one steel ball is pushed horizontally while a second is released straight down, at the exact same instant.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1** State a testable claim about whether a dropped ball and a horizontally launched ball, released from the same height at the same instant, hit the ground at the same time.
- PE2** Draw on prior evidence and experience when reasoning about the claim — and recognize where everyday intuition can be misleading.

- PE3** **Observe the launcher–drop device** and record what actually happens when both balls are released together.
- PE4** **Sketch velocity vectors at regular time intervals** for the dropped ball (A), the launched ball (B), and a hypothetical zero-gravity case (C).
- PE5** **Explain, using vector reasoning, why horizontal and vertical motion are independent** — constant v_x , ever-changing v_y .
- PE6** **Write the launched ball's velocity as a vector**, $\vec{v}(t) = v_{0x} \hat{i} + (-9.8t) \hat{j}$, and identify what each term represents.

Evidence: what you already have

Before you commit to a claim, take stock of what you've already seen — even if it turns out to be misleading. That's part of building good evidence:



Two coins off a table

Flick one coin off the edge of a table while a friend drops a second coin from the same height, at the same instant. Which do you expect to hit the floor first?



Movie physics

In action movies, a car driven off a cliff often seems to hang in the air before it drops. Is that really how gravity works?



A thrown ball

You've watched a ball thrown sideways arc downward through the air. Does it really spend more time falling than one just dropped from the same height?

In this note, we work through

- I. What's your **claim**: do the two balls hit the ground at the same time, or does one beat the other?
- II. What **evidence** or experience do you already have that's relevant — and can that intuition be wrong?
- III. What do you actually **observe** when both balls are released from the launcher–drop device at the same instant?
- IV. How do the velocity vectors on trajectories A, B, and C **explain** what's happening in each direction?

V. Why does the hypothetical no-gravity path C look the way it does?

Claim — predict first: a ball is dropped from a tower at the same instant a second ball is fired horizontally from the same height. Ignoring air resistance, which one hits the ground first?

The dropped ball lands first.

The launched ball lands first.

They land at exactly the same time.

It depends on how fast the ball is launched.

Observe: the launcher–drop device

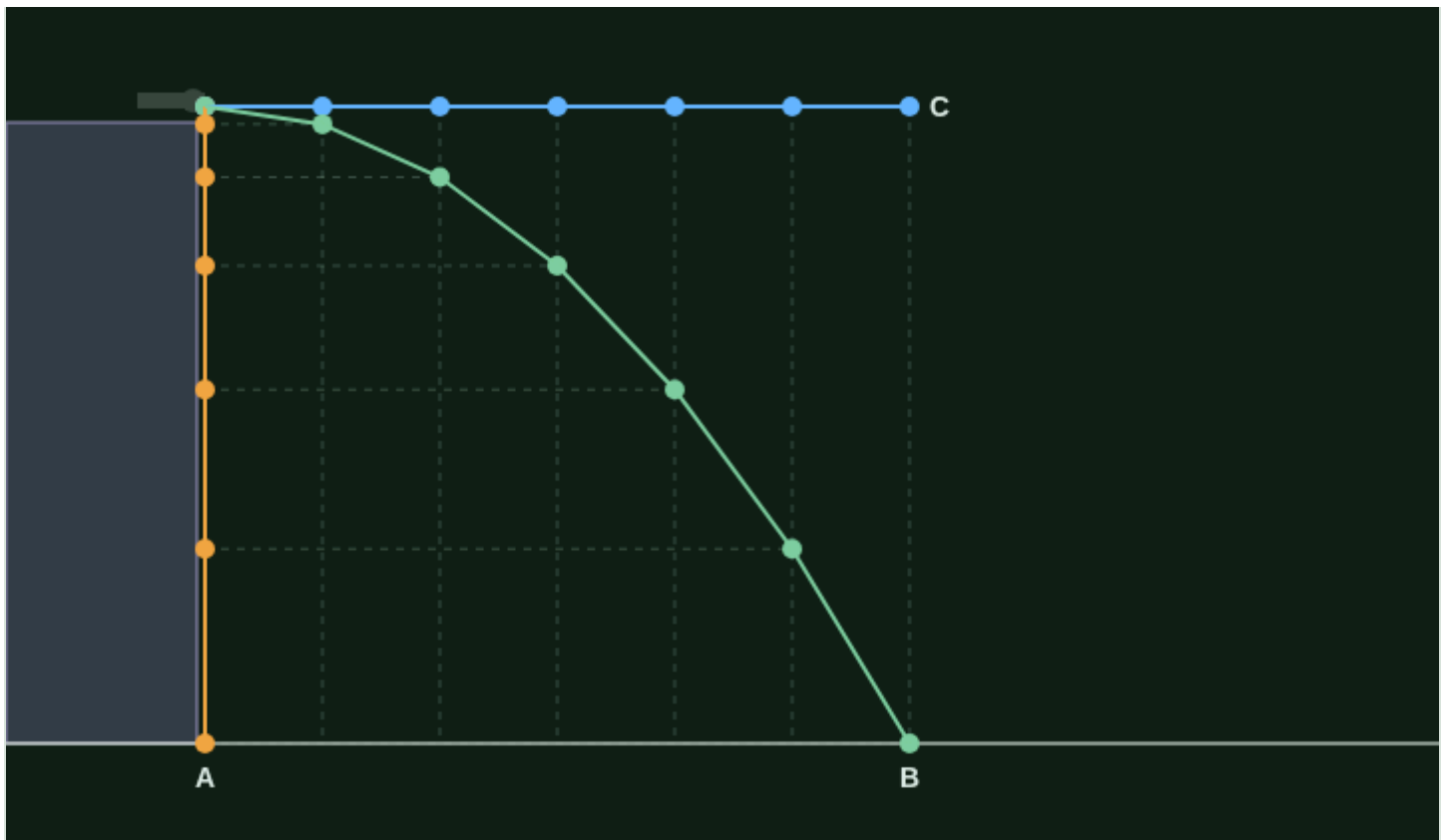
The device pictured above holds two steel balls at the same height. A spring-loaded arm pushes one ball out horizontally at the exact instant it releases the second ball to drop straight down — no reaction time, no guessing about "at the same time." Watch (and listen) as your instructor fires it.

What you should notice: however hard the launcher fires the horizontal ball, you hear a single "thud" when the balls land — not two. Both balls reach the floor at the same instant, every time, no matter the launch speed.

Play with it: trajectories A, B, and C

Trajectory **A** is the dropped ball, **B** is the horizontally launched ball, and **C** is a hypothetical trajectory for the launched ball if it were fired in a gravity-free region of space. Press play and watch the shared clock — notice which two balls always reach the bottom together, and which one never does. Try the launch-speed slider, then press play again.

► INTERACTIVE · FREEFALL VS. HORIZONTAL LAUNCH



Launch speed v_{0x}

12.0 m/s



► Play

Show velocity vectors

↺ Reset

t = 0.00 s

$v_x \approx -$

$v_y \approx -$

Notice: A and B always land at the same instant — only C, with no gravity, keeps flying past the edge of the diagram.



Explain: reading the vectors

Every arrow you toggled on above is a velocity vector, and together they tell the whole story. Path A's arrow only ever points straight down, and it keeps growing — that's gravity acting alone, accelerating the ball downward at 9.8 m/s^2 with nothing pushing it sideways. Path C's arrow only ever points sideways, and it never changes length — that's what horizontal motion looks like with no gravity at all: constant velocity, forever. Path B's arrow is just those two combined, added component by component — and the dashed guides show it directly: B's horizontal piece exactly

matches C at every instant, and B's vertical piece exactly matches A at every instant. Neither component knows the other exists.

That's the **independence of motion**: a projectile's horizontal velocity and vertical velocity change (or don't change) completely independently of each other. For the horizontally launched ball, we can write that as a single vector equation:

$$\vec{V}(t) = v_{0x} \hat{i} + (-9.8 t) \hat{j}$$

The horizontal component, v_{0x} , never appears with a t — it's constant. The vertical component, $-9.8t$, grows steadily with time — the same acceleration acting on the dropped ball. Set v_{0x} to zero and this is exactly Path A's velocity. That's the reasoning behind the claim: since vertical motion (the only thing that determines *when* a ball reaches the ground) is identical for A and B, they must land at the same instant — no matter how fast the ball is launched sideways.

Watch: working through the note

WALKTHROUGH VIDEO • ~10 MIN

Video embed goes here — print the guided notes below before you press play.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. The idea that a projectile's horizontal and vertical motions don't affect each other is called...

- ☐ Newton's third law.
- ☐ The independence of motion.
- ☐ Conservation of momentum.
- ☐ Terminal velocity.

Q2. In a CER (Claim–Evidence–Reasoning) response, the "Reasoning" part should...

- ☐ Restate the claim in different words.
- ☐ List observations without interpreting them.
- ☐ Connect the evidence to the claim using a scientific principle.
- ☐ Predict a new, unrelated experiment.

Q3. A ball is dropped from a 20 m tower at the same instant a second ball is fired horizontally from the same height at 15 m/s. Ignoring air resistance, how long does it take the launched ball to hit the ground? (Hint: which velocity component actually controls this?)

s

Q4. In $\vec{v}(t) = v_{0x} \hat{i} + (-9.8t) \hat{j}$ for the launched ball, the term $-9.8t$ represents...

- ☐ The constant horizontal velocity.
- ☐ The increasing downward vertical velocity due to gravity.

☐ The total speed of the ball.

☐ The horizontal position at time t .

Check my answers

Clear

Your notes

PRINTABLE VERSION

Read this note (PDF)

A static, printable copy of everything on this page — good for offline reading or printing.

AFTER THE MODULE

Completed notes (PDF)

Released when Module 2 closes — check your work against it.